PV Hemorrhage in Pregnancy

Focused History
- Gestation,
- Gravidity / Parity
- Past obstetric history
- LMP
- Previous PV bleeds?
- Last fetal movements
- Include postural BP assessment

Assess amount, rate and nature of PV bleeding

Apply absorbent pad to perineum area
Avoid PV examination

Early pregnancy hemorrhage
<20 weeks' gestation

Antepartum hemorrhage
>20 weeks' gestation

Hemodynamically Stable?
Heart Rate <100bpm
Blood Pressure >90/60mmHg

YES

- Place absorbent pad to perineum area
- Calm and reassure mother
- Ensure mother is in a position of comfort (left lateral tilt if possible)

NO

Consider ALS Back-Up

Position mother in the **left lateral tilt position**

Administer Oxygen
To maintain Sp02 >94-98%

Administer Normal Saline
500-1ml NS IV/IO boluses
(max 2000ml)

Administer TXA
1g IV/IO over 10 minutes

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Consider Analgesia

Go to Pain CPG

If significant nausea,
Consider Metoclopramide
10mg IV/IO/IM
For IV/IO: administer as infusion in 100ml NaCl over 10 minutes

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Patient Care Goals

1. Recognize serious conditions associated with hemorrhage during pregnancy, even when the cause is not apparent (e.g., ectopic pregnancy, abruptio placenta, placenta previa).
2. Provide adequate resuscitation for hypovolemia and stabilize the patient.
3. Ensure rapid transport to an appropriate receiving facility to address potential fetal compromise and maternal shock.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Female patient with vaginal bleeding in any trimester.
- Female patient with pelvic pain or possible ectopic pregnancy.
- Consider pregnancy in any female between the ages of 10-60 years.

Exclusion Criteria

- Childbirth and active labor (See Childbirth CPG)
- Postpartum hemorrhage (See Post-Partum Hemorrhage CPG)

Key Patient Assessment Points

1. **Determine Obstetric / Gynecologic History**
   - Gestation
   - Gravidity/parity
   - Past obstetric history, last menstrual period (LMP)
   - Previous PV bleeds, last fetal movements
2. **Estimation of Blood Loss & Bleeding Severity**
   - Quantify vaginal bleeding (e.g., number of soaked pads, passing clots >5 cm, or estimated loss >500 mL)
   - Assess the rate PV bleeding.
3. **Evaluate for Signs of Shock or Instability**
   - Assess for signs of shock:
     - Evaluate mental status (e.g., confusion, agitation, or lethargy).
     - Note pallor, sweating, or clammy skin, dizziness, weakness, or palpitations.
     - Assess heart rate (>100 bpm), or blood pressure (<90/60 mmHg) are indicative of significant blood loss.
     - Postural blood pressure assessment (A drop in BP >20 mmHg, and/or a rise in heart rate between lying and standing measurements indicates significant hypovolemia.
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - Estimated blood loss >500mL
       - Ongoing bleeding with signs of shock

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
   - Determine obstetric history.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **General Management**
   - Place absorbent pad to perineum.
   - Calm and reassure mother.
   - Ensure mother is in a position of comfort (left lateral tilt if possible).
4. **Administer high-flow oxygen**
   - At 10-15 Lpm via non-rebreather (NRB) mask.
5. **Establish large-bore IV access**.

- Consider second large bore IV if patient is hemodynamically unstable.

6. **If Hemodynamically Unstable:**
   - Administer IV Normal Saline 500 ml boluses (max 2000 ml).
7. **Consider Tranexamic acid (TXA):**
   - Indications:
     - Profuse bleeding, with signs of shock or/
     - Significant shock symptoms
   - Dose:1 gram IV, diluted in 100ml NaCl, administered over 10 minutes.
8. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Patients in the third trimester of pregnancy should be transported on their left side or with the uterus manually displaced to the left if hypotensive.
- Do not place hand/fingers into the vagina to stop bleeding.

Clinical Insights & Additional Guidance

- Syncope
  - Syncope can be a presenting symptom of intra-abdominal hemorrhage from ectopic pregnancy or antepartum hemorrhage (e.g., spontaneous abortion, placental abruption, placenta previa).
  - Postural BP assessment is a critical indicator of hypovolemia.
- Avoid unnecessary vaginal manipulation to prevent worsening hemorrhage.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Symptoms such as vaginal bleeding, pelvic pain, syncope, or suspected ectopic pregnancy.
- Injuries / Information: Obstetric/gynecologic history including gestation, gravidity, parity, last menstrual period (LMP), past obstetric history, previous PV bleeds, and last fetal movements.
- Signs:
  - Vital signs (BP, HR, RR, SpO_2), temperature, mental status (AVPU/GCS).
  - Estimated blood loss (e.g., number of soaked pads, clot size >5 cm, or volume).
  - Rate of PV bleeding.
  - Signs of hypovolemia or shock (e.g., pallor, clammy skin, tachycardia >100 bpm, hypotension <90/60 mmHg, postural BP drop >20 mmHg).
- Treatment:
  - IV/IO access placement and fluid resuscitation details
  - Oxygen administration, Tranexamic acid (TXA) administration.
  - Patient positioning
- Allergies: All known allergies, especially to medications (e.g., TXA).
- Medications: Current or recent use of relevant drugs (e.g., anticoagulants).
- Background: Chronic illnesses (e.g., hypertension, diabetes), recent obstetric complications, or risk factors for ectopic pregnancy or placental issues.
- Other:
  - Dispatch and hospital coordination notes (e.g., notification of PV hemorrhage and shock status).
  - Document any deterioration or red flag conditions (e.g., syncope, severe hypotension).

Key Performance Indicators

1. Administer IV fluids for all hemodynamically unstable patients.

References

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